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<tr><td>LESSON PLAN</td><td>COURSE: MHF4U</td></tr>
<tr><td>Date: Lesson #13</td><td>Unit of Study: Chapter 7 Tools and Strategies for Solving Exponential and Logarithmic Equations</td></tr>
<tr><td>Materials / Resources:

- Advanced Function 12 Study Guide
- Chapter 6 & 7 Formula, Aid and Homework Sheet(s)
- KWL Chart</td><td>Lesson Focus

- Techniques for Solving Exponential Equations
- Product and Quotient Laws
- Techniques for Solving Logarithmic Equations</td></tr>
<tr><td colspan="2">Prior Knowledge:

- Able to solve for simple exponential equations
- Graphs of logarithms & their characteristics (roots especially)</td></tr>
<tr><td colspan="2">Overall Expectations:

1. Demonstrate an understanding of the relationship between exponential expressions and logarithmic expressions, evaluate logarithms, and apply the laws of logarithms to simplify numeric expressions
2. Identify and describe some key features of the graphs of logarithmic functions, make connections among the numeric, graphical, and algebraic representations of logarithmic functions, and solve related problems graphically
3. Solve exponential and simple logarithmic equations in one variable algebraically, including those in problems arising from real-world applications</td></tr>
<tr><td colspan="2">Specific Expectations:

1.4 Make connections between the laws of exponents and the laws of logarithms [e.g., use the statement $10^{a+b} = 10^{a}10^{b}$ to reduce that $x + y = (xy)$], verify the laws of logarithms with or without technology (e.g., use patterning to verify the quotient law for logarithms by evaluating expression such as $1000 - 100$ and then rewriting the answer as a logarithmic term to the same base), and use the laws of logarithms to simplify and evaluate numerical expressions
3.1 Recognize equivalent algebraic expressions involving logarithms and exponents, and simply expressions of these types
3.2 Solve exponential equations in one variable by determining a common base (e.g., solve $4^{x} = 8^{x+3}$ by expressing each side as a power of 2) and by using logarithms (e.g., solve $4^{x} = 8^{x+3}$ by taking the logarithm base 2 of both sides), recognizing that logarithms base 10 are commonly used (e.g., solving $3^{x} = 7$ by taking the logarithm base 10 of both sides)
3.3 Solve simple logarithmic equations in one variable algebraically [e.g., $(5x + 6) = 2$, $(x + 1) = 1$]
3.4 Solve problems involving exponential and logarithmic equations algebraically, including problems arising from real-world applications</td></tr>
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Learning Goals:

- Students will know how to solve equations involving exponential expressions using a variety of methods (change of base, algebraic manipulation, use of graphing technology)
- Students will be able to identify and reject extraneous roots to equations involving exponential expressions
- Students will know how to solve problems involving exponential equations
- Students will be able to apply to the product and quotient laws of logarithms
- Students will be able to determine restrictions on variables for expressions involving exponents logarithms
- Students will know how to solve for equations involving logarithmic expressions using a variety of methods (rewrite numbers as logarithms, apply laws of logarithms, use of graphing technology)
- Students will have knowledge of solving problems involving logarithmic equations

Success Criteria:

1. By the end of this lesson I will be able to solve equations involving exponential expressions using a variety of methods (change of base, algebraic manipulation, use of graphing technology)
2. By the end of this lesson I will be able to identify and reject extraneous roots to equations involving exponential expressions
3. By the end of this lesson I will be able to solve problems involving exponential equations
4. By the end of this lesson I will be able to apply the product and quotient laws of logarithms
5. By the end of this lesson I will be able to state restrictions on variables for expressions involving exponents and logarithms
6. By the end of this lesson I will be able to solve equations involving logarithmic expressions using a variety of methods (rewrite numbers as logarithms, apply laws of logarithms, use of graphing technology)
7. By the end of this lesson I will be able to solve problems involving logarithmic equations

Assessment Strategies:

<u>Formative</u>: test #4

<u>Summative</u>: homework for chapters 7.2, 7.3 & 7.4

Time: **Part 1**	**Introduction:** Take up / check homework from lesson #12 and check for student's understanding. If needed, go over specific questions.
Time: **Part 2**	**Lesson:** <u>Chapter 7.2 Techniques for Solving Exponential Equations</u> Explain the concept of applying logarithm to both sides of an equation and how equality is still maintained. Using the power laws of logarithms, solve for exponential equations using these laws and applying logarithms to the equations. Identify the meaning behind solving for exponential equations. Much like solving for a quadratic or

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<tr><td></td><td>polynomial equation, a root is much be round to satisfy the equation when $y = 0$. Identify extraneous roots and recognize under what circumstances they occur in.

<u>Chapter 7.3 Product and Quotient Laws of Logarithm</u>
Using the formula sheets, introduce the product and quotient laws of logarithm and apply proofs to justify these laws. Apply the laws to evaluate and solve logarithmic equations. Relate these two laws to exponents and how we can deduce them from exponential equations. (e.g. form $10^{a+b} = 10^{a}10^{b}$ to $x + y = (xy)$). Identify restrictions for the laws.
<u>Chapter 7.4 Techniques for Solving Logarithmic Equations</u>
Using methods such as factoring from quadratic equations, solve for logarithmic equations using these methods to extract the root. Identify the meaning behind solving for logarithmic equations. Much like solving for a quadratic or polynomial equation, a root is much be round to satisfy the equation when $y = 0$. Identify extraneous roots and recognize under what circumstances they occur in.</td></tr>
<tr><td>Time:

Part 3</td><td>Wrap-up:

Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions. Use KWL chart to better understand student’s learning needs for chapter #6 and chapter #7.</td></tr>
<tr><td colspan="2">Notes, reminders & homework:

<u>Reminder</u>:
- Quiz #5 (to be written before / after current class)
- Test #4 (to be written before / after lesson #14)

<u>Homework</u>
Chapter 7.2: #1 – 10, 14, 15
Chapter 7.3: #1, 3, 6, 7, 13, 16
Chapter 7.4: #1 – 6, 8, 10, 11</td></tr>
<tr><td colspan="2">Lesson Reflection:</td></tr>
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